

Volume-Weighted Momentum (VWMOM) Alpha Factor

Factor Research Report (Sample)

1. Factor Definition

VWMOM measures recent price momentum weighted by relative trading volume. For instrument i on day t : $VWMOM_{i,t} = \sum_{k=1}^L (V_{i,t-k+1} / VMA_{i,t}) * r_{i,t-k+1}$, where r is the daily return, V is daily volume, VMA is the 20-day volume moving average, and $L = 10$ lookback days. The factor is cross-sectionally z-scored each day.

2. Economic Rationale

Returns accompanied by above-average volume carry more information. Weighting momentum by relative volume emphasizes price moves backed by genuine participation and filters low-liquidity noise.

3. Data and Universe

Daily OHLCV bars from the qlib CN dataset. Universe: CSI300 constituents. Frequency: daily, close-to-close. Neutralization: cross-sectional z-score; optional industry neutralization.

4. Backtest Settings

Top/bottom decile long-short portfolio, daily rebalance, 5 basis points one-way cost. Metrics: annualized return, information ratio, max drawdown, factor IC and rank IC.

5. Expected Behavior

Positive rank IC around 0.02-0.04 on A-shares in trending regimes; decay half-life roughly 5 trading days. Monitor turnover, which is moderate due to the 10-day window.